

RobustBench Model Ranking Stability under AutoAttac

Machine Learning security

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01 Goal of the project

- Evaluate 5 CIFAR-10 robust models from RobustBench
- Test them under l_∞ adversarial attacks using AutoAttack

Vary perturbation strength ϵ :

- $1/255, 4/255, 8/255, 12/255, 16/255$

Compare:

- Robust accuracy
- Model ranking stability

02 The Current Limitation

Fixed Benchmark:

The standard RobustBench leaderboard currently assumes a fixed perturbation strength of $\epsilon = 8/255$.

Dynamic Vulnerability:

In real-world scenarios, adversarial robustness is not static and may change significantly depending on the value of ϵ

The Core Research Question

Ranking Stability:

Do model rankings remain stable across different levels of ϵ , or does the "best" model change as attack strength increases?

ROBUSTBENCH

The Gold Standard Leaderboard

- **Systematic Tracking:** Centralized platform to monitor global progress in adversarial robustness.
- **Fair Evaluation:** Researchers compare models under identical, standardized conditions.
- **Scientific Rigor:** Solves the "Overestimation Problem" by replacing inconsistent testing with comparable, reliable results.

AUTOATTACK

The Evaluation Engine

- **Ensemble Defense:** Combines strong white-box and black-box attacks to eliminate "weak attack" bias.
- **Fully Automated:** Reliable and reproducible results with zero manual tuning required.
- **Standard Metric:** The primary method used by RobustBench to prevent "fake" or exaggerated robustness.

Key elements of memorable presentation



SELECTED BENCHMARKS

- **Carmon2019Unlabeled:** Leverages unlabeled data to boost robustness.
- **Rebuffi2021Fixing:** High-capacity model; most stable across all ϵ values.
- **Rice2020Overfitting:** Manages the trade-off between clean and robust accuracy.
- **Engstrom2019Robustness:** Standard baseline for l_∞ adversarial training.
- **Wu2020Adversarial:** Mid-tier model; consistent under strong perturbations.

Note: All are CIFAR-10 l_∞ adversarially trained models.

CORE COMMONALITIES

- **DATASET**
ALL MODELS WERE TRAINED AND TESTED ON CIFAR-10
- **THREAT MODEL**
SPECIFICALLY DESIGNED FOR l_∞ ADVERSARIAL TRAINING.
- **SOURCE**
PRETRAINED WEIGHTS WERE RETRIEVED DIRECTLY FROM THE ROBUSTBENCH REPOSITORY

Experimental setup

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01 DATASET

CIFAR-10 (subset 50–100 samples)

02 ATTACK

AutoAttack (Standard version), including APGD-CE and APGD-DLR

03 THREAT MODEL

l_∞ bounded perturbations with $\varepsilon \in \{1, 4, 8, 12, 16\} / 255$.

04 DEVICE

NVIDIA Tesla T4 GPU (Google Colab environment)

05 METRIC

Robust Accuracy (percentage of adversarial samples correctly classified).

The Engines of AutoAttack

08

APGD-CE & APGD-DLR

01

What is APGD? (The "Auto" Advantage)

Definition: APGD stands for Auto-PGD, an advanced version of the classic Projected Gradient Descent (PGD) attack.

The Upgrade: It features automatic step tuning, which removes the need for manual parameter adjustments.

Benefit: This ensures a more reliable and consistent attack across different model architectures.

02

Dual-Loss Strategy

AutoAttack utilizes two distinct loss functions to prevent models from "cheating" the evaluation:

APGD-CE (Cross-Entropy):

- Targets the standard loss used in most training.
- Effective at breaking models with standard gradient-based vulnerabilities.

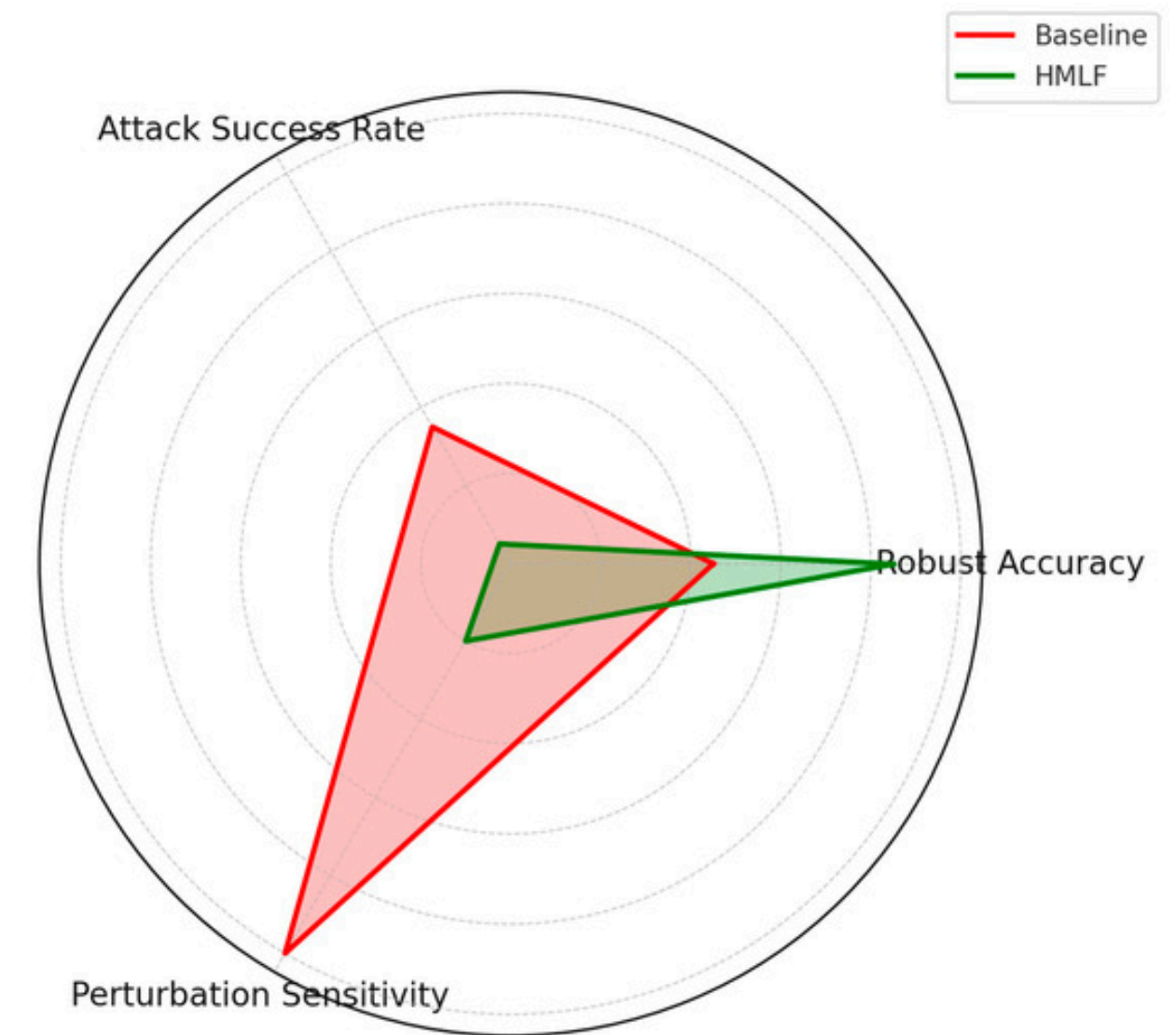
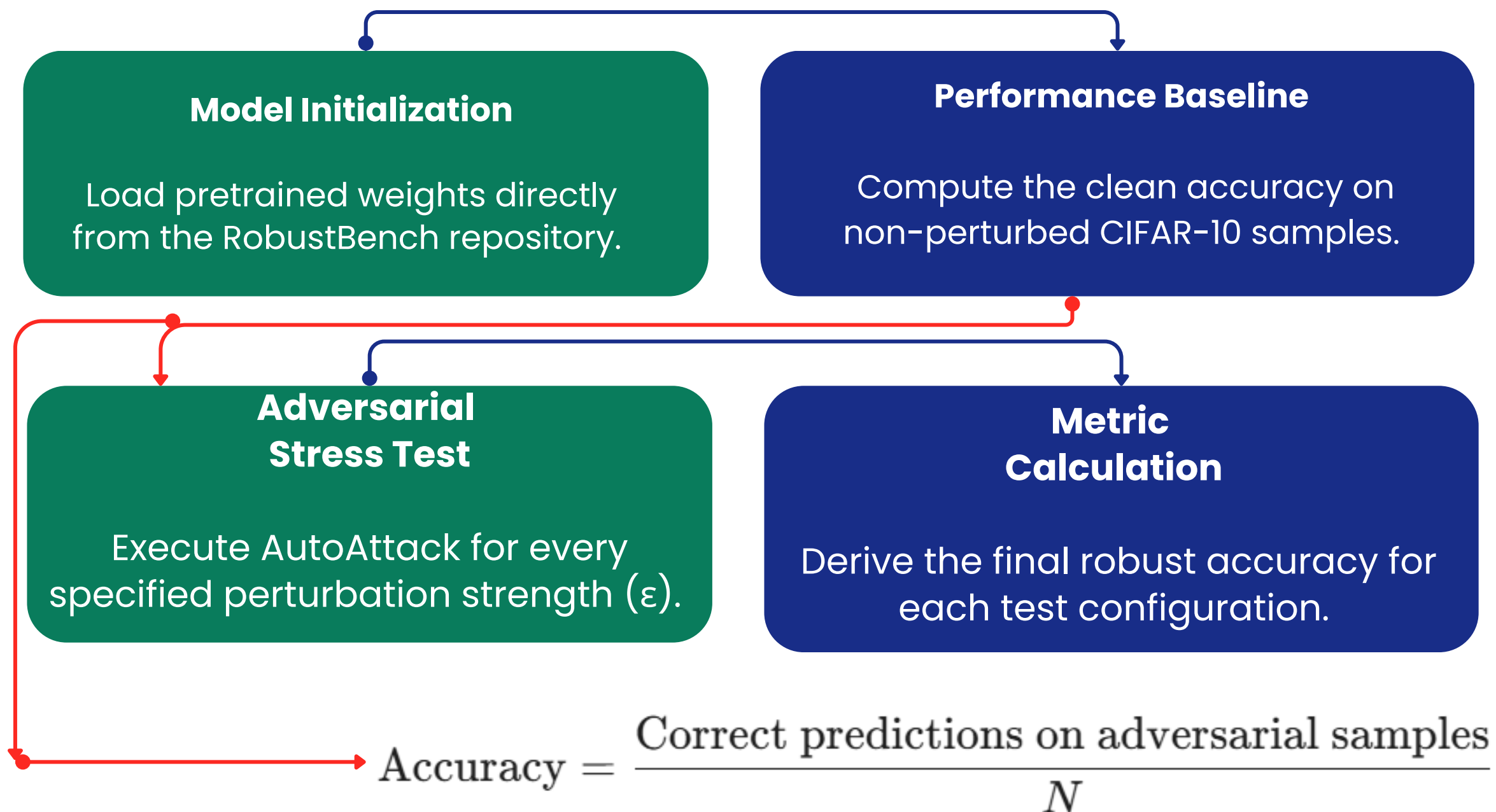
APGD-DLR (Difference of Logits Ratio):

- Uses DLR loss, which is shift-invariant and scale-invariant.
- Goal: To break models that use "**gradient masking**" or other tricks to resist standard CE attacks.

Methodology & Ranking analysis

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To evaluate the stability of model rankings, the following systematic approach was applied to each selected architecture:



```
=====
Loading model: Rebuffi2021Fixing_70_16_cutmix_extra
Downloading models/cifar10/Linf/Rebuffi2021Fixing_70_16_cutmix_extra.pt (gdrive_id=1qKDTp6IJ18UXZaRtbYuo_t0tuDl_4mLg).
Downloading...
From (original): https://drive.google.com/uc?id=1qKDTp6IJ18UXZaRtbYuo\_t0tuDl\_4mLg
From (redirected): https://drive.google.com/uc?id=1qKDTp6IJ18UXZaRtbYuo\_t0tuDl\_4mLg&confirm=t&uuid=713abb2f-99bf-4241-9fb0-737f837505e2
To: /content/models/cifar10/Linf/Rebuffi2021Fixing_70_16_cutmix_extra.pt
100%|██████████| 1.07G/1.07G [00:26<00:00, 40.7MB/s]
Clean accuracy: 0.96
epsilon = 1/255
setting parameters for rand version
using rand version including apgd-ce, apgd-dlr.
initial accuracy: 96.00%
apgd-ce - 1/1 - 0 out of 48 successfully perturbed
robust accuracy after APGD-CE: 96.00% (total time 2117.5 s)
apgd-dlr - 1/1 - 0 out of 48 successfully perturbed
robust accuracy after APGD-DLR: 96.00% (total time 4226.2 s)
max Linf perturbation: 0.00000, nan in tensor: 0, max: 1.00000, min: 0.00000
robust accuracy: 96.00%
epsilon = 4/255
setting parameters for rand version
using rand version including apgd-ce, apgd-dlr.
initial accuracy: 96.00%
apgd-ce - 1/1 - 3 out of 48 successfully perturbed
robust accuracy after APGD-CE: 90.00% (total time 2111.7 s)
apgd-dlr - 1/1 - 3 out of 45 successfully perturbed
robust accuracy after APGD-DLR: 84.00% (total time 4164.0 s)
max Linf perturbation: 0.01569, nan in tensor: 0, max: 1.00000, min: 0.00000
robust accuracy: 84.00%
epsilon = 8/255
setting parameters for rand version
using rand version including apgd-ce, apgd-dlr.
initial accuracy: 96.00%
apgd-ce - 1/1 - 17 out of 48 successfully perturbed
robust accuracy after APGD-CE: 62.00% (total time 2107.1 s)
apgd-dlr - 1/1 - 2 out of 31 successfully perturbed
robust accuracy after APGD-DLR: 58.00% (total time 3515.5 s)
max Linf perturbation: 0.03137, nan in tensor: 0, max: 1.00000, min: 0.00000
robust accuracy: 58.00%
epsilon = 12/255
setting parameters for rand version
using rand version including apgd-ce, apgd-dlr.
initial accuracy: 96.00%
apgd-ce - 1/1 - 24 out of 48 successfully perturbed
robust accuracy after APGD-CE: 48.00% (total time 2113.2 s)
apgd-dlr - 1/1 - 1 out of 24 successfully perturbed
robust accuracy after APGD-DLR: 46.00% (total time 3397.1 s)
max Linf perturbation: 0.04706, nan in tensor: 0, max: 1.00000, min: 0.00000
robust accuracy: 46.00%
epsilon = 16/255
setting parameters for rand version
using rand version including apgd-ce, apgd-dlr.
initial accuracy: 96.00%
```

```
To: /content/models/cifar10/Linf/Carmon2019Unlabeled.pt
100%|██████████| 153M/153M [00:06<00:00, 24.8MB/s]
Clean accuracy: 0.92
epsilon = 1/255
setting parameters for rand version
using rand version including apgd-ce, apgd-dlr.
initial accuracy: 92.00%
apgd-ce - 1/1 - 1 out of 46 successfully perturbed
robust accuracy after APGD-CE: 90.00% (total time 280.4 s)
apgd-dlr - 1/1 - 0 out of 45 successfully perturbed
robust accuracy after APGD-DLR: 90.00% (total time 563.2 s)
max Linf perturbation: 0.00392, nan in tensor: 0, max: 1.00000, min: 0.00000
robust accuracy: 90.00%
epsilon = 4/255
setting parameters for rand version
using rand version including apgd-ce, apgd-dlr.
initial accuracy: 92.00%
apgd-ce - 1/1 - 9 out of 46 successfully perturbed
robust accuracy after APGD-CE: 74.00% (total time 284.8 s)
apgd-dlr - 1/1 - 1 out of 37 successfully perturbed
robust accuracy after APGD-DLR: 72.00% (total time 545.5 s)
max Linf perturbation: 0.01569, nan in tensor: 0, max: 1.00000, min: 0.00000
robust accuracy: 72.00%
epsilon = 8/255
setting parameters for rand version
using rand version including apgd-ce, apgd-dlr.
initial accuracy: 92.00%
apgd-ce - 1/1 - 20 out of 46 successfully perturbed
robust accuracy after APGD-CE: 52.00% (total time 286.0 s)
apgd-dlr - 1/1 - 0 out of 26 successfully perturbed
robust accuracy after APGD-DLR: 52.00% (total time 472.5 s)
max Linf perturbation: 0.03137, nan in tensor: 0, max: 1.00000, min: 0.00000
robust accuracy: 52.00%
epsilon = 12/255
setting parameters for rand version
using rand version including apgd-ce, apgd-dlr.
initial accuracy: 92.00%
apgd-ce - 1/1 - 26 out of 46 successfully perturbed
robust accuracy after APGD-CE: 40.00% (total time 286.3 s)
apgd-dlr - 1/1 - 0 out of 20 successfully perturbed
robust accuracy after APGD-DLR: 40.00% (total time 456.3 s)
max Linf perturbation: 0.04706, nan in tensor: 0, max: 1.00000, min: 0.00000
robust accuracy: 40.00%
epsilon = 16/255
setting parameters for rand version
using rand version including apgd-ce, apgd-dlr.
initial accuracy: 92.00%
apgd-ce - 1/1 - 35 out of 46 successfully perturbed
robust accuracy after APGD-CE: 22.00% (total time 284.3 s)
apgd-dlr - 1/1 - 1 out of 11 successfully perturbed
robust accuracy after APGD-DLR: 20.00% (total time 389.2 s)
max Linf perturbation: 0.06275, nan in tensor: 0, max: 1.00000, min: 0.00000
robust accuracy: 20.00%
```



```

... Running on: cuda
100%|██████████| 170M/170M [00:04<00:00, 42.0MB/s]

=====
Loading model: Rice2020overfitting
Downloading models/cifar10/Linf/Rice2020overfitting.pt (gdrive_id=1vC_Twazji7lBjeMQvAD9uEQxi9Nx2oG-).
Downloading...
From (original): https://drive.google.com/uc?id=1vC\_Twazji7lBjeMQvAD9uEQxi9Nx2oG-
From (redirected): https://drive.google.com/uc?id=1vC\_Twazji7lBjeMQvAD9uEQxi9Nx2oG-&confirm=t&uuid=d4c507f3-ac1c-435a-ab60-ce1efa1daf93
To: /content/models/cifar10/Linf/Rice2020overfitting.pt
100%|██████████| 738M/738M [00:09<00:00, 79.1MB/s]
Clean accuracy: 0.86
epsilon = 1/255
setting parameters for rand version
using rand version including apgd-ce, apgd-dlr.
initial accuracy: 86.00%
apgd-ce - 1/1 - 1 out of 43 successfully perturbed
robust accuracy after APGD-CE: 84.00% (total time 1111.2 s)
apgd-dlr - 1/1 - 0 out of 42 successfully perturbed
robust accuracy after APGD-DLR: 84.00% (total time 2229.3 s)
max Linf perturbation: 0.00392, nan in tensor: 0, max: 1.00000, min: 0.00000
robust accuracy: 84.00%

```

ROBUST ACCURACY TABLE:

	epsilon	1	4	8	12	16	
	model						
	Rice2020Overfitting	0.84	0.76	0.52	0.32	0.14	

... Running on: cuda

=====

Loading model: Engstrom2019Robustness

Downloading models/cifar10/Linf/Engstrom2019Robustness.pt (gdrive_id=1etqmQsksNIWBvBQ4r8ZFk_3FJ1LWr8Rr).

Downloading...

From: https://drive.google.com/uc?id=1etqmQsksNIWBvBQ4r8ZFk_3FJ1LWr8Rr

To: /content/models/cifar10/Linf/Engstrom2019Robustness.pt

100%|██████████| 94.4M/94.4M [00:06<00:00, 13.9MB/s]

Clean accuracy: 0.9

epsilon = 1/255

setting parameters for rand version

using rand version including apgd-ce, apgd-dlr.

initial accuracy: 90.00%

apgd-ce - 1/1 - 2 out of 45 successfully perturbed

robust accuracy after APGD-CE: 86.00% (total time 184.3 s)

apgd-dlr - 1/1 - 0 out of 43 successfully perturbed

robust accuracy after APGD-DLR: 86.00% (total time 372.2 s)

max Linf perturbation: 0.00392, nan in tensor: 0, max: 1.00000, min: 0.00000

robust accuracy: 86.00%

ROBUST ACCURACY TABLE:

	epsilon	1	4	8	12	16
model						
Engstrom2019Robustness		0.86	0.76	0.48	0.24	0.12



... Running on: cuda

=====

Loading model: Wu2020Adversarial

Downloading models/cifar10/Linf/Wu2020Adversarial.pt (gdrive_id=13LBcgNvhFppCFG22i1xATrahFPfMgXGf).

Downloading...

From (original): <https://drive.google.com/uc?id=13LBcgNvhFppCFG22i1xATrahFPfMgXGf>

From (redirected): <https://drive.google.com/uc?id=13LBcgNvhFppCFG22i1xATrahFPfMgXGf&confirm=t&uuid=6f4faff1-fcd8-4827-ae12-f8df1027eb7f>

To: /content/models/cifar10/Linf/Wu2020Adversarial.pt

100%|██████████| 185M/185M [00:03<00:00, 46.4MB/s]

Clean accuracy: 0.88

epsilon = 1/255

setting parameters for rand version

using rand version including apgd-ce, apgd-dlr.

initial accuracy: 88.00%

apgd-ce - 1/1 - 2 out of 44 successfully perturbed

robust accuracy after APGD-CE: 84.00% (total time 362.9 s)

apgd-dlr - 1/1 - 0 out of 42 successfully perturbed

robust accuracy after APGD-DLR: 84.00% (total time 717.2 s)

max Linf perturbation: 0.00392, nan in tensor: 0, max: 1.00000, min: 0.00000

robust accuracy: 84.00%

ROBUST accuracy: 18.00%

ROBUST ACCURACY TABLE:

epsilon	1	4	8	12	16
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model					
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Wu2020Adversarial	0.84	0.7	0.5	0.36	0.18
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Models defending

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Rebuffi2021Fixing_70_16_cutmix_extra

- “The most robust and stable model across all epsilon values, maintaining the highest adversarial accuracy.”

Carmon2019Unlabeled

- “A balanced and stable model whose robustness degrades gradually under stronger perturbations.”

Wu2020Adversarial

- “A consistent mid-performing model that becomes comparatively stronger at higher epsilon values.”

Rice2020Overfitting

- “A model with strong low-epsilon performance but reduced stability under stronger adversarial attacks.”

Engstrom2019Robustness

- “A model that performs well under weak perturbations but experiences the fastest robustness degradation as epsilon increases.”

Final results

Model	Clean	1/255	4/255	8/255	12/255	16/255
Rebuffi2021Fixing_70_16_cutmix_extra	96%	96%	84%	58%	46%	~32–34%
Carmon2019Unlabeled	92%	90%	72%	52%	40%	20%
Wu2020Adversarial	88%	84%	70%	50%	36%	18%
Rice2020Overfitting	86%	84%	76%	52%	32%	14%
Engstrom2019Robustness	90%	86%	76%	48%	24%	12%

Key Observations

Observation 1:

Universal Degradation: All models experience non-linear robust accuracy decay as ε increases.

Observation 2:

Stability Leader: Rebuffi2021 maintains the most gradual slope, retaining ~33% robustness even at extreme perturbation levels.

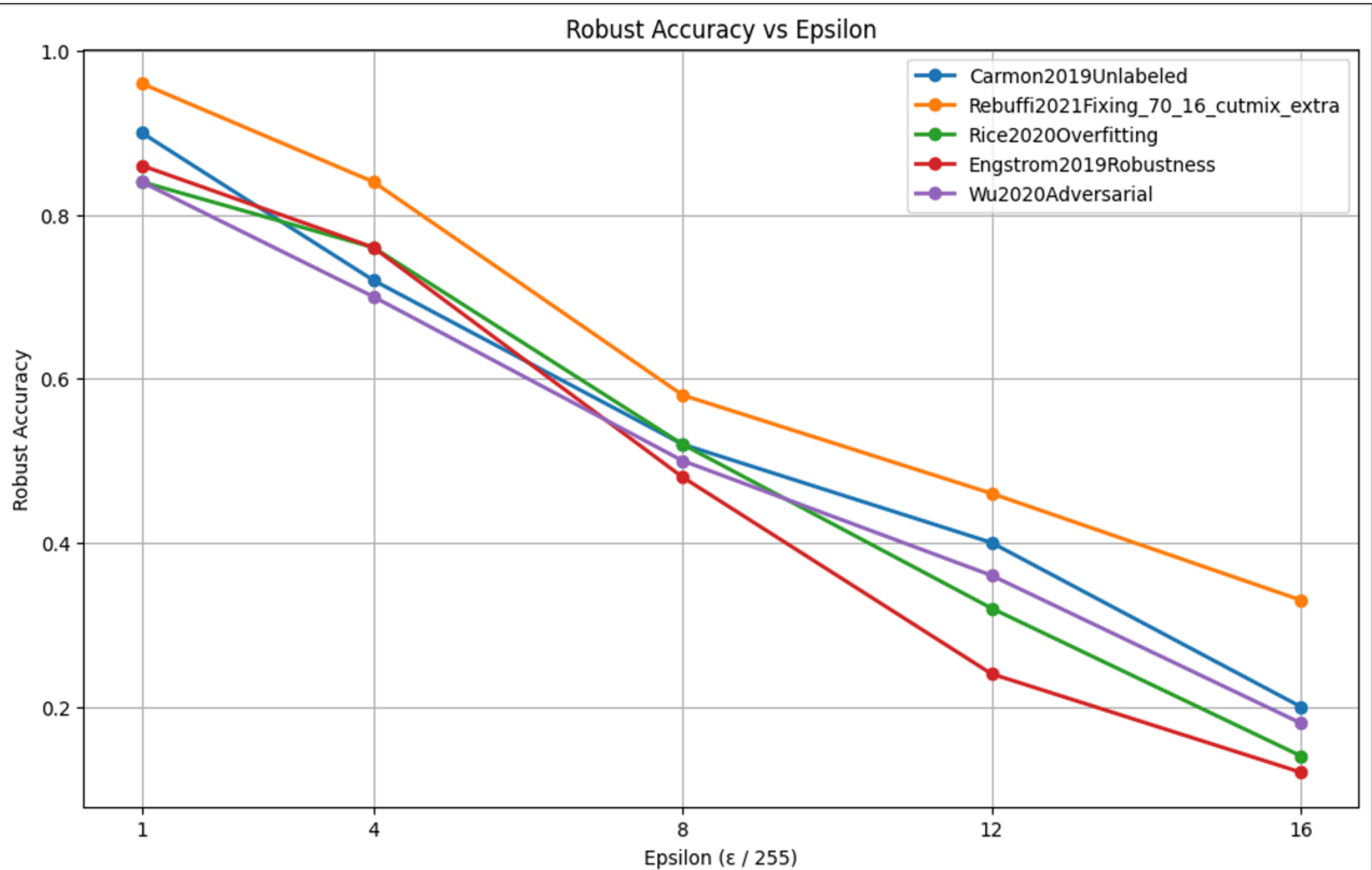
Ranking Stability Analysis

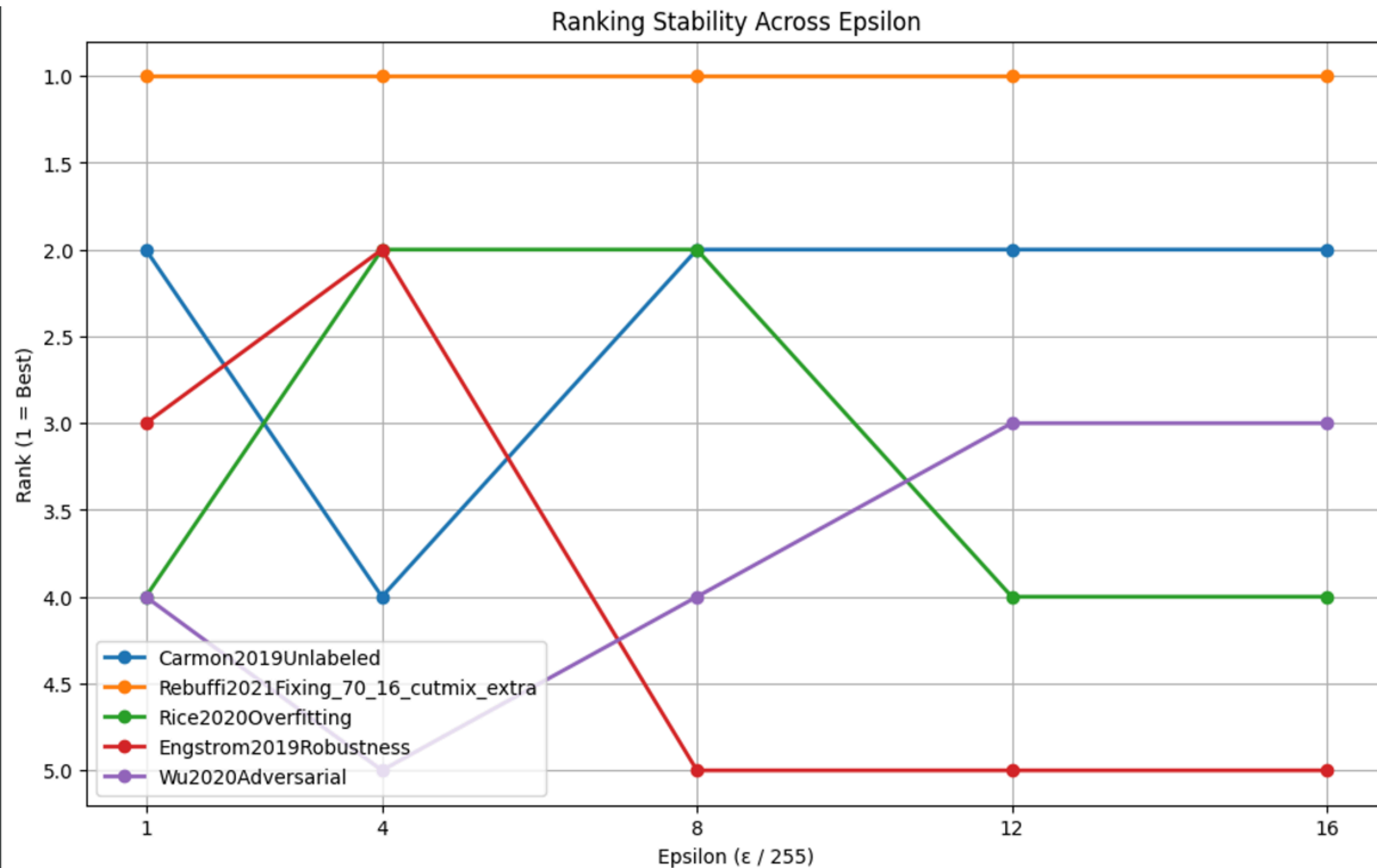
The Leaderboard "Flip"

Model rankings are not static. A model that ranks higher at $\varepsilon = 1/255$ (e.g., Rice2020) may be overtaken by others as ε increases.

Stability Insight

Evaluating robustness at a single ε value (the "Standard Benchmark") fails to capture the full defensive profile of neural networks.





Conclusion

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This study evaluated five CIFAR-10 l_∞ robust models from RobustBench using AutoAttack across multiple perturbation strengths ranging from $1/255$ to $16/255$. The results showed a consistent decrease in robust accuracy as epsilon increased, confirming the sensitivity of all models to stronger adversarial perturbations.

Among the evaluated models, Rebuffi2021Fixing_70_16_cutmix_extra achieved the best overall performance and demonstrated the highest robustness stability across all epsilon values. In contrast, several mid-ranked models changed positions as epsilon increased, indicating that robustness rankings are not fully stable and depend strongly on the chosen attack strength.

These findings highlight an important limitation of robustness leaderboards: model rankings can vary significantly under different evaluation settings. Therefore, adversarial robustness should be assessed across multiple epsilon values rather than relying on a single benchmark configuration.

**Thank
You!**